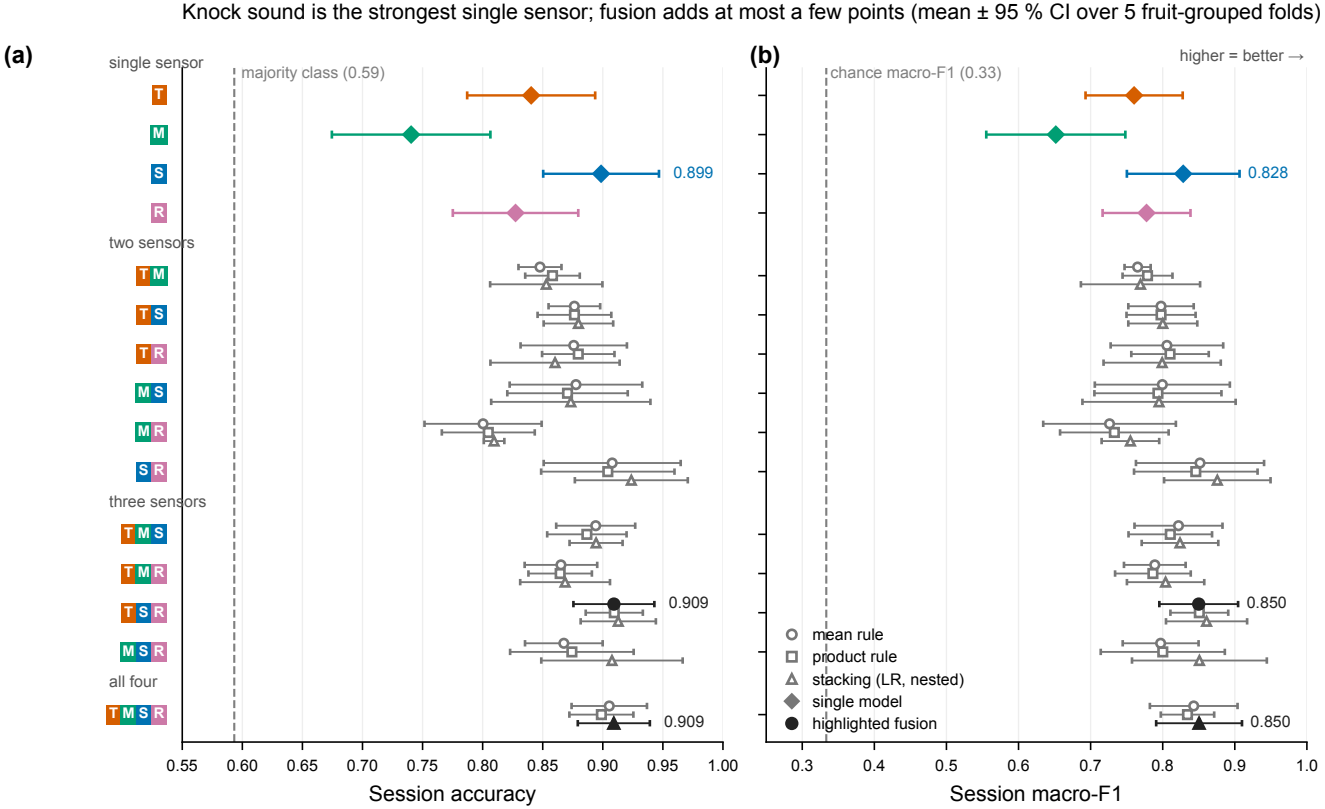
Knock sound is the strongest single sensor; fusion adds at most a few points (mean ± 95 % CI over 5 fruit-grouped folds)
(a)
(b)
single sensor
two sensors
three sensors
all four
T
M
S
R
majority class (0.59)
chance macro-F1 (0.33)
higher = better →
0.899
0.828
0.909
0.850
0.909
0.850
mean rule
product rule
stacking (LR, nested)
single model
highlighted fusion
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
0.95
1.00
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Session accuracy
Session macro-F1